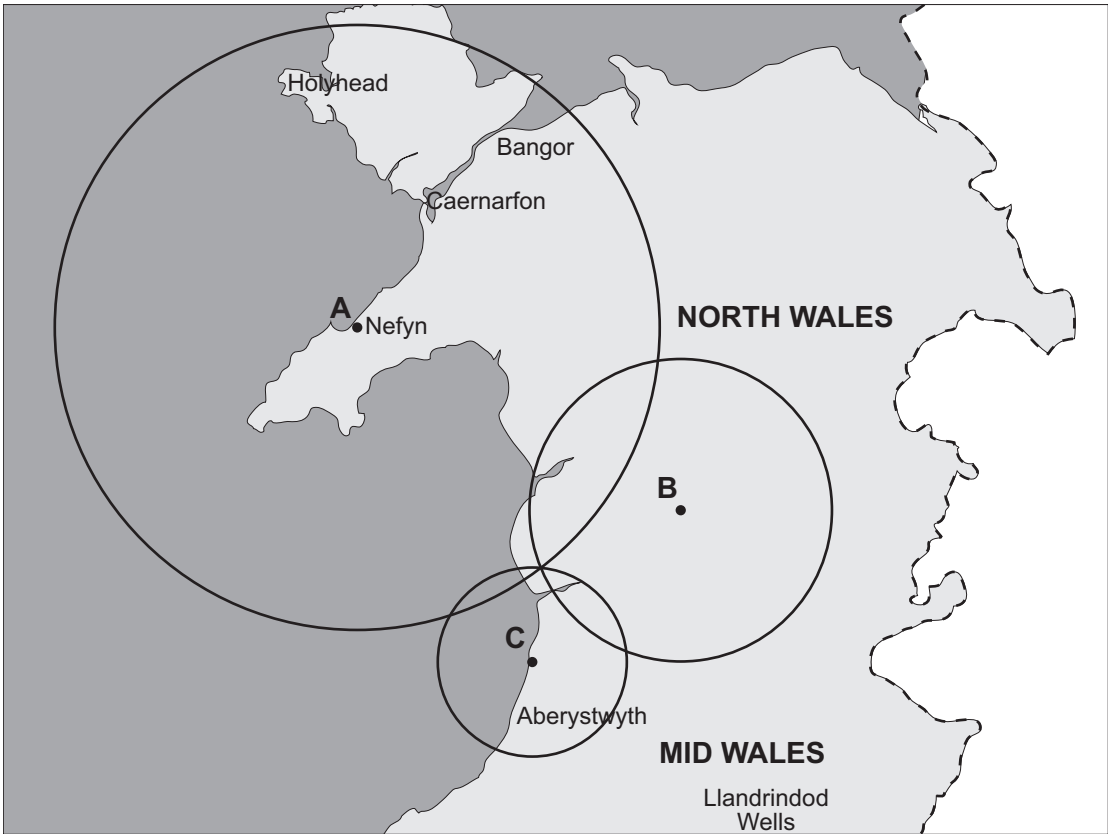


Answer all questions.

1. The waves from an earthquake can be monitored by seismological stations located on the surface of the Earth.



The diagram shows circles drawn around three seismological stations **A**, **B** and **C** in Mid and North Wales. The radius of each circle shows the distance of the epicentre from each station.

- (a) (i) **Label on the diagram** with a cross (X) the location of the epicentre. [1]
- (ii) Explain why at least three seismological stations are needed to locate the epicentre. [2]

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- (b) Explain which station (**A**, **B** or **C**) detected the earthquake first. [2]

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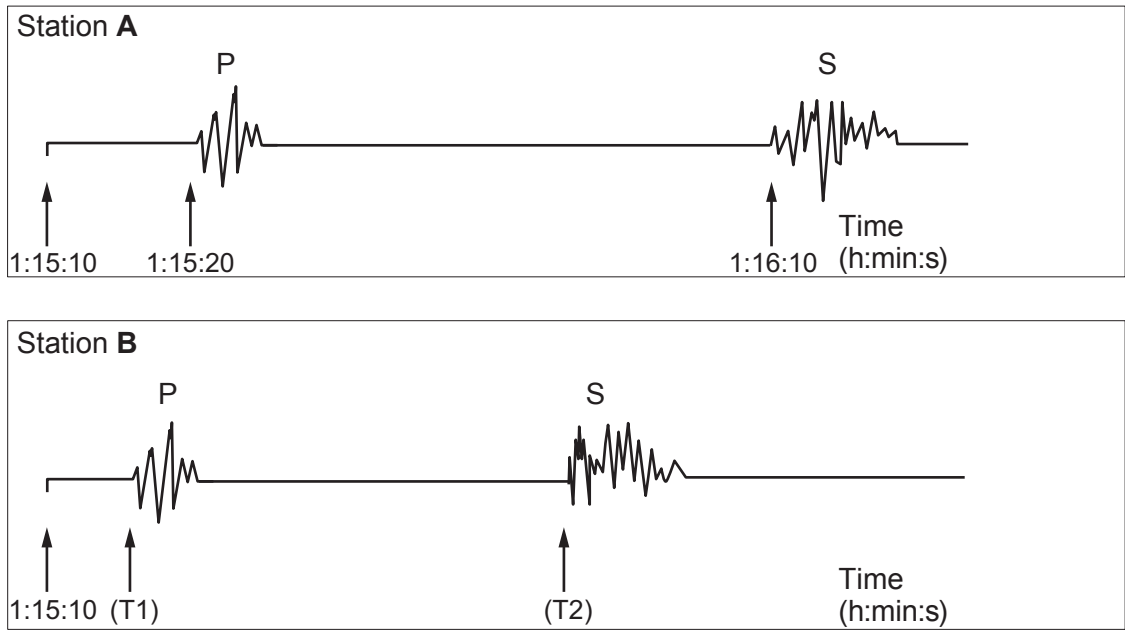
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- (c) The diameter of the circle around station A **is double** the diameter of the circle around station B.

The earthquake happened at 1 hour : 15 minutes : 10 seconds (1:15:10) in the morning.

The following signals were obtained at stations **A** and **B**.



Due to an error on the equipment the **arrival times** (T1 and T2) for the two signals were not recorded at station **B**.

Using signal information from station **A**, determine the times that the P waves and the S waves arrived at station **B**. Explain how you arrived at your answers. [3]

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Arrival time of P waves (T1) = hours minutes seconds.

Arrival time of S waves (T2) = hours minutes seconds.